Experiment No-11 : Map Coloring Using CSP

Aim

To write a Python program for Map Coloring using a Constraint Satisfaction Problem (CSP).

Programming Code:

```
def is_valid(region, color, assignment, neighbours):
    for neighbour in neighbours[region]:
        if assignment.get(neighbour) == color:
            return False
    return True

def map_coloring(regions, colors, neighbours, assignment=None):
    if assignment is None:
        assignment = {}

    if len(assignment) == len(regions):
        return assignment

    region = next(r for r in regions if r not in assignment)
    for color in colors:
        if is_valid(region, color, assignment, neighbours):
            assignment[region] = color
            result = map_coloring(
                regions, colors, neighbours, assignment)
            if result:
                return result
            del assignment[region]

    return None

regions = ['A', 'B', 'C', 'D']
colors = ['Red', 'Green', 'Blue']
neighbours = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}

solution = map_coloring(regions, colors, neighbours)
print("Map Coloring using CSP")
if solution:
    for region in regions:
        print(region, ":", solution[region])
else:
    print("No solution exists")
```

OUTPUT:

```
Map Coloring using CSP
A : Red
B : Green
C : Blue
D : Red
```

Result

Thus, the Map Coloring problem was implemented successfully using CSP. The program assigns colors to regions while ensuring that neighboring regions do not have the same color.